

SCOTTSDALE, AZ, USA

WMO: 722789

Lat: 33.623N

Lon: 111.911W

Elev: 1473

StdP: 13.93

Time Zone: -7.00 (W07)

Period: 01-19

WBAN: 03192

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	36.1	38.8	2.0	6.4	63.0	6.5	8.1	65.2	16.7	55.6	13.4	57.2	1.0	50	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	108.6	68.6	106.6	68.3	104.8	68.1	74.4	91.4	73.5	91.7	72.9	92.1	7.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.5	118.7	80.3	68.6	111.0	82.2	66.5	102.9	84.6	38.8	90.9	38.1	91.3	37.4	91.3	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.8	12.6	10.8	DB	30.9	112.7	2.6	2.2	29.0	114.3	27.5	115.6	26.0	116.9	24.1	118.5	(4)
(5)			WB	26.3	76.1	2.8	0.8	24.3	76.7	22.7	77.2	21.1	77.7	19.1	78.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.6	55.2	57.5	64.6	71.1	79.0	89.7	93.6	92.0	86.6	75.0	63.6	53.8	(6)	
	DBStd	15.21	5.63	6.28	6.62	5.95	6.41	4.86	4.10	4.27	5.16	6.22	6.71	5.69	(7)	
	HDD50	63	20	11	1	0	0	0	0	0	0	0	2	29	(8)	
	HDD65	1086	305	217	89	13	1	0	0	0	0	7	103	348	(9)	
	CDD50	8661	180	220	453	632	899	1191	1353	1303	1098	775	410	147	(10)	
	CDD65	4210	1	6	76	198	435	741	888	838	648	317	61	1	(11)	
	CDH74	64050	34	145	992	2378	5989	12291	14850	13659	9546	3518	631	17	(12)	
(13)	CDH80	40250	1	13	285	951	3307	8326	10444	9301	5863	1608	151	0	(13)	
(14)	Wind	WSAvg	3.6	2.5	2.9	3.6	4.5	4.6	4.3	4.9	4.2	3.8	3.1	2.7	2.4	(14)
(15) Precipitation	PrecAvg	12.2	1.7	1.5	1.5	0.5	0.2	0.1	1.1	1.7	1.0	0.8	0.8	1.4	(15)	
	PrecMax	24.4	8.6	5.6	5.2	1.9	2.1	0.7	3.2	4.6	4.3	5.6	3.5	4.8	(16)	
	PrecMin	5.4	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.4	0.0	0.0	0.0	0.1	(17)	
	PrecStd	4.4	1.9	1.5	1.2	0.5	0.4	0.1	0.8	0.9	0.9	1.0	0.7	1.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.0	81.9	91.0	95.7	103.5	111.3	111.1	110.2	106.2	98.3	88.6	76.1	(19)	
		MCWB	51.9	54.0	58.0	58.7	62.9	65.9	69.4	69.5	68.5	63.5	58.9	52.8	(20)	
	2%	DB	74.1	78.7	86.4	92.1	99.9	107.9	108.8	107.7	103.3	94.2	84.0	72.6	(21)	
		MCWB	51.2	53.0	56.3	57.8	62.0	65.4	69.6	69.9	67.7	62.4	56.8	51.9	(22)	
	5%	DB	71.2	75.0	82.9	88.8	97.3	105.9	106.8	105.6	101.0	91.5	81.0	69.5	(23)	
		MCWB	50.4	51.5	54.9	56.7	60.7	65.3	69.5	69.7	67.1	61.5	56.1	50.9	(24)	
	10%	DB	67.9	71.4	79.8	85.7	94.3	103.3	104.6	102.9	98.9	89.0	77.6	65.8	(25)	
		MCWB	49.3	50.5	53.8	55.5	59.6	64.4	69.4	69.6	67.0	60.7	54.7	49.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.2	58.1	59.8	61.4	66.5	71.5	75.5	75.4	74.1	70.5	62.2	58.3	(27)	
		MCDB	65.4	69.5	82.7	88.3	91.5	96.4	91.0	91.2	88.2	80.0	75.1	66.1	(28)	
	2%	WB	55.1	56.1	58.0	59.4	64.3	69.8	74.4	74.5	72.8	67.6	59.7	55.5	(29)	
		MCDB	63.9	67.6	80.9	87.1	91.7	97.2	91.6	91.4	88.8	84.2	75.8	66.3	(30)	
	5%	WB	53.2	54.4	56.5	57.9	62.3	68.1	73.6	73.7	71.6	65.2	58.0	53.3	(31)	
		MCDB	64.3	67.1	78.1	84.1	93.2	98.2	92.2	91.7	89.4	83.4	75.4	63.5	(32)	
	10%	WB	51.4	52.7	55.0	56.7	60.6	66.4	73.0	73.0	70.6	63.3	56.4	51.3	(33)	
		MCDB	63.8	67.0	75.3	82.0	91.8	98.5	92.8	92.1	89.7	83.1	74.2	62.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	22.0	21.7	23.7	23.6	23.7	24.5	20.1	19.9	21.0	22.6	22.4	20.6	(35)	
		MCDBR	27.4	28.0	28.4	28.0	27.4	26.5	22.1	22.6	23.3	25.6	26.7	25.7	(36)	
	5% WB	MCWBR	11.7	11.1	9.3	9.2	8.1	7.3	5.7	5.4	6.3	8.1	9.9	11.2	(37)	
		MCWBR	19.9	20.7	25.5	24.6	24.9	21.6	18.9	18.5	17.4	18.5	21.2	19.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.280	0.286	0.313	0.322	0.331	0.341	0.402	0.389	0.344	0.321	0.289	0.280	(40)	
		taud	2.574	2.570	2.472	2.435	2.442	2.414	2.371	2.396	2.496	2.510	2.554	2.579	(41)	
	Ebn at Noon		301	309	306	306	302	298	279	282	292	292	294	294	(42)	
		Edh at Noon	25	28	33	36	36	37	39	37	32	29	25	23	(43)	
(44) All-Sky Solar Radiation	RadAvg		1063	1363	1840	2295	2577	2680	2329	2147	1930	1551	1183	962	(44)	
	RadStd		83	110	94	69	96	100	109	74	73	94	52	53	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+347	+176	(46)
(47)	Regional (15 neighbors)	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	-146	+178	+61	(47)

Nomenclature: See separate page